

# Sound and Video



YJ40201: Fundamentals of Programming  
Stony Brook Institute at Anhui University

# Basic sound playback

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- **How to play a sound file?**
  - **Step 1. Import the Processing sound library.**
  - **Step 2. Declare a SoundFile object.**
  - **Step 3. Initialize the SoundFile object.**
  - **Step 4. Start to play the sound.**

# Example: Simple sound playback

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```
import processing.sound.*;

SoundFile song;

void setup() {
    size(640, 360);
    song = new SoundFile(this,
                          "groove.mp3");
    song.play();
}

void draw() {
}

void mousePressed() {
    if (song.isPlaying()) {
        song.pause();
    } else {
        song.play();
    }
}
```

# More methods of SoundFile

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- **loop( )**
- **stop( )**
- **amp( )**
- **rate( )**
- **pan( )**

# Live Video

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- **How to capture and display the video image?**
  - **Step 1. Import the Processing video library.**
  - **Step 2. Declare a Capture object.**
  - **Step 3. Initialize the Capture object.**
  - **Step 4. Start to capture the image.**
  - **Step 5. Read the image from the camera.**
  - **Step 6. Display the video image.**

## Example: Display video

```
import processing.video.*;

Capture video;

void setup() {
    size(640, 480);
    video = new Capture(this,
                        640, 480);
    video.start();
}

void captureEvent(Capture video) {
    video.read();
}

void draw() {
    image(video, 0, 0);
}
```

# Recorded Video

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- **Displaying recorded video**
  - **Step 1. Declare a Movie object.**
  - **Step 2. Initialize Movie object.**
  - **Step 3. Start movie playing.**
  - **Step 4. Read frame from movie.**
  - **Step 5. Display the movie.**

# Example: Display movie

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```
import processing.video.*;

Movie movie;

void setup() {
    size(320, 240);
    movie = new Movie(this,
                       "cat.mov");
    movie.play();
}

void movieEvent(Movie movie)
{
    movie.read();
}

void draw() {
    image(movie, 0, 0);
}
```